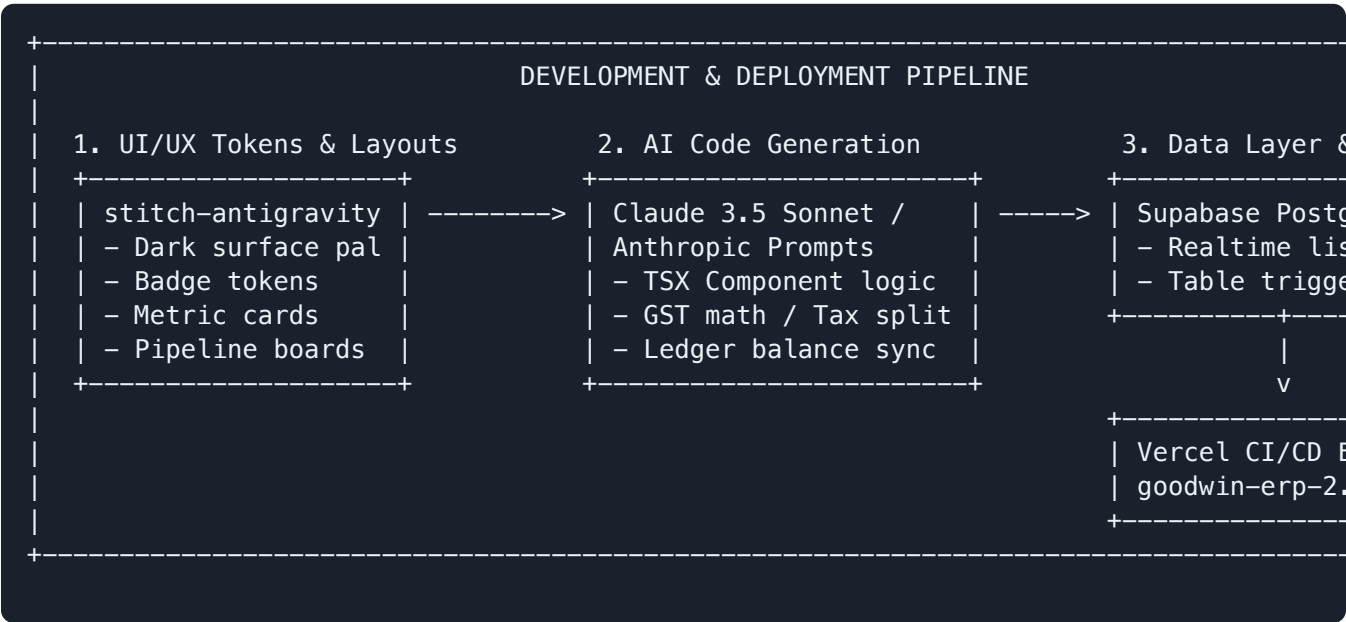


Standard Operating Procedure (SOP)

Document ID: SOP-GW-2026-V1
System Target: Goodwin Batteries ERP, Ledger-Pro & HRMS Enterprise Platform
URL: goodwin-erp-2.vercel.app
Architecture: Next.js / TypeScript + stitch-antigravity UI + Supabase PostgreSQL + Vercel Edge
Effective Date: September 2026

1. System Architecture & Build Methodology



1.1 Technology Stack & Development Lifecycle

- stitch-antigravity (UI/UX Scaffolding):** Established the foundational dark mode aesthetic (deep-slate base `#0d1117` / `#161b22`) and functional token system (Green for Active/Paid, Amber for Pending, Red for Overdue) across administrative tables, analytics summaries, and CRM boards.
- Prompt Engineering & Claude (Anthropic):** Acted as the core AI developer. Domain prompts supplied to Claude generated TypeScript functional components, database schema definitions, automated 18% GST calculations (9% CGST + 9% SGST), party ledger debit/credit balances, and drag-and-drop mechanics.
- Supabase (Managed PostgreSQL & Real-Time Sync):** Managed backend service configured at `https://fyxqyy1clamwigvdzjem.supabase.co` . Provides persistent relational storage, foreign key integrity, and active WebSocket listeners that push live updates across ERP, Accounting, and CRM modules without browser refreshes.
- Vercel:** Edge-optimized hosting connected to the Git repository for continuous

integration and automated production releases.

2. Core ERP Operations Guide

2.1 Customer & Dealer Management

- **Directory Access:** Navigate to **Customers & Dealers** to audit partner accounts, Unique Origin Identifiers (UOIs, e.g., `GW-CUST-1007`), phone numbers, GSTIN credentials, and live outstanding balances.
- **Adding Accounts:** Click **+ Add Customer**, fill in the legal entity name, state, phone, email, and tax identification, then submit to initialize their account ledger at ₹0.
- **Actions:** Use **Edit** to modify party details or **Ledger** to review transactional statements.

2.2 Supplier & Vendor Management

- Navigate to **Suppliers & Vendors** to oversee component, container, and raw lead plate manufacturers.
- Click **+ Add Vendor/Supplier** to register new suppliers with contact details and credit limits.
- Monitor procurement liability through the **Payable Amount** column.

2.3 Products & Inventory Catalog

- Navigate to **Products & Inventory** to review product lines, SKU codes, HSN numbers (e.g., `8507` for lead-acid accumulators), purchase rates, retail prices, on-hand shelf counts, and warranty duration terms.
- Click **+ Add Products** to register new battery models (e.g., *Endura N150 - 12V 150Ah*) and assign them to physical bin locations (e.g., *Warehouse A/Rack-01*).

2.4 Sales & GST Invoicing

1. Navigate to **Sales/GST Invoices** and select **Create Sales Invoice** (or **+ New Invoice / Order**).
2. Select customer, product items, and quantity.
3. The system automatically computes base taxable figures, applies 18% GST, and renders the grand total.
4. Click **Save & Generate Invoice** to automatically deduct warehouse stock, post the debit balance to the dealer ledger, and prepare a printable tax invoice.

2.5 Purchases & Purchase Orders (POs)

- Navigate to **Purchases & POs** and click **+ Create Purchases & POs**.
- Select the vendor, log component batches (lead plates, containers, acid), taxable value, and GST input credits.
- Click **View PO** to review procurement status.

2.6 Returns (Credit & Debit Notes)

- Navigate to **Returns (Credit/Debit)** and click **+ Create Return**.
- Select return classification:
 - **Credit Note (Sales Return)**: Issued to customers for returned or damaged goods.
 - **Debit Note (Purchase Return)**: Issued to manufacturers for rejected raw components.
- Provide line amounts and failure causes (e.g., *Manufacturing defect in terminal/cell*). Approval immediately updates balances and physical inventory.

2.7 Battery Warranty Tracker

- **Registration**: Go to **Battery Warranty** and click **Register Battery Warranty**. Enter the battery serial number (e.g., **GW-458674**), customer ID, product model, and sale date. The system sets the warranty expiration date automatically.
- **Claims Processing**: Search the serial number during customer return inquiries. Click **Process Claim** to initiate diagnostics, repairs, or replacements, or click **View Card** to access the digital warranty certificate.

3. Ledger-Pro: Financial Accounting Operations

3.1 Financial Monitoring

- Review liquid assets directly on the **Ledger-Pro Dashboard**: Cash in Hand, Bank Balances (e.g., HDFC Current Account), aggregate Party Receivables, and Supplier Payables.
- Review cash flow velocity across daily, weekly, and monthly tracking intervals.

3.2 Parties Directory & Invoices

- The **Parties Directory** aggregates both customers and vendors into a single accounting list showing active balances and credit limits.
- The **Sales/Invoices** screen tracks settled and unsettled invoice values.

3.3 Recording Payments

- **Payment In (Collections)**: Go to **Payment In** → click **+ Take Payment In**. Choose the

customer, input the payment method (Bank/Cash), record the transaction reference (e.g., **UTR-88991122**), and enter the collected amount. Submitting immediately reconciles the customer balance.

- **Payment Out (Disbursements):** Go to **Payment Out** → click **+ Take Payment Out**. Choose the supplier, transaction type, reference UTR, and amount to settle vendor liabilities.

4. CRM & Lead Management System

4.1 Lead Directory & Tracking

- Open **Lead Management > Leads** and click **+ Add Lead** to record incoming prospects, estimated deal value, and source attribution (e.g., WhatsApp, Referral).

4.2 Visual Pipeline (Kanban)

- Navigate to **Pipeline** to track opportunities across defined stages: **NEW** → **CONTACTED** → **FOLLOW-UP** → **QUALIFIED** → **WON / LOST** .
- Drag cards across stage columns to update opportunity status and trigger lead alerts.

4.3 Follow-up Action Center

- Review scheduled interactions filed under **Overdue**, **Today**, and **Upcoming**.
- Use quick-action buttons (**Call**, **WhatsApp**, **Open Lead**) to launch customer communication directly.

5. Goodwin HRMS (Human Resources Management)

- **Employee Roster:** Manage personnel profiles, company email accounts, department assignments, and job titles under **HRMS > Employees**.
- **Projects & Tasks:** Create operational goals, log billable project hours, and assign tasks under **Projects & Tasks**.
- **Today's Attendance:** Open **Attendance** to record daily employee status (**Present**, **Absent**, **Late**, **Half Day**). Select any employee profile to inspect their monthly attendance grid.
- **Leave Approvals:** Review incoming employee time-off applications under **Leave** and mark them as **Approved** or **Rejected**.
- **Payroll Execution:** Open **Payroll** and select **+ Process Payroll** at month-end to reconcile attendance logs, apply leave deductions, and generate net pay distributions.

6. Cloud Database & Configuration Maintenance

Configuration Area	Operational Requirement	Action Steps
Supabase Cloud Connection	Maintain live database sync and verify operational state.	Go to Settings & Cloud DB and click Test Connection (Ping) to ensure real-time synchronization is active.
Schema Migrations	Deploy new table definitions, columns, and foreign keys.	Click Copy SQL Schema , open the Supabase SQL Editor, paste the SQL query, and execute.
Battery Voltage & Capacity	Define selectable attributes across product and quoting forms.	Update comma-separated lists (e.g., <code>12V, 24V, 48V</code> and <code>8Ah, 14Ah, 35Ah, 150Ah</code>) in the Battery Configurator.
Warehouse Logistics	Maintain physical inventory rack and bin locations.	Enter storage identifiers line-by-line (e.g., <code>Warehouse A/Rack-01</code>) and click Save Configuration Settings .
Sales Rep Assignments	Assign agents to CRM leads and sales routes.	Add agent codes (e.g., <code>RJ, AJ</code>) to populate sales assignment selectors across the system.